ecbench report: ECBS1h24a44d1bc573

Label: chain-free Koblitz IC: half-dimension Frobenius-invariant base, m = 2 (lookup oracle), K_0 / F_2^31

Description: LEADS.md L2. The chaining term (m-2)n vanishes at m = 2 when the Frobenius-invariant base has dimension near n/2. Arms at n = 31: the strong signed-Frobenius rho reference and its A/A control; ic.pipeline on koblitz-standard-subspace bases of dimension 10, 12, 14 with the m = 2 pair-table oracle (negation fold). Counted operations; every phase charged; a measurement, not a performance claim. Dimension 16 was removed after its warm-up failed a 36.5 GB allocation (sessions/n31-interrupted-d16); the m = 3 one-link inherited-F4 arm was removed after it timed out at 2400 s on the first target at 34 MiB (sessions/n31-m3-timeout) and is measured at n = 23 instead (spec_n23.json).

Generated by `ecbench report` (ecbench.report/v1). Observations only; derived numbers carry the key of their derivation (last section).

Session

field	value
session id	ECBS1h24a44d1bc573
spec id	ECS1h10090e558038
commit	5221aeeba08bd2a5f10ce47cdc5c74e0960e0020 (dirty)
status	complete
run options	cpus="none", lock=/tmp/crypto-bench.lock, wait_for_lock=false, allow_busy=true

Host and isolation

Intel(R) Xeon(R) Processor @ 2.10GHz; 4 logical / 4 physical cores; kernel 6.18.44-fc-v139; 15.7 GiB; env class ECBENV2he7f30f001a0e; virtualization cpu_flag=hypervisor, systemd_detect_virt_vm=kvm.

Isolation earned by the 40 measured records: L0 40, L1 0, L2 0, L3 0. Required of a wall-clock figure: L0.

blocker	measured records
L1: no CPU reservation (–cpus none, or no affinity control on this OS)	40

Arms

arm	role	method	params	expected S [d4]
rho-strong	reference	rho.signed_frobenius_strong	dp_bits=4; lanes=32; step_cap_factor=2000	0.1592
rho-strong-aa	control	rho.signed_frobenius_strong	dp_bits=4; lanes=32; step_cap_factor=2000	0.1592
ic-m2-d10	candidate	ic.pipeline	factor_base=koblitz-standard-subspace:dimension=10; linalg=incremental-gauss; max_trials=2000000; oracle=pdp3-koblitz:m=2,engine=rootfind-negfold; solver=; solver_budget_seconds=0; targets=walk	–
ic-m2-d12	candidate	ic.pipeline	factor_base=koblitz-standard-subspace:dimension=12; linalg=incremental-gauss; max_trials=2000000; oracle=pdp3-koblitz:m=2,engine=rootfind-negfold; solver=; solver_budget_seconds=0; targets=walk	–
ic-m2-d14	candidate	ic.pipeline	factor_base=koblitz-standard-subspace:dimension=14; linalg=incremental-gauss; max_trials=2000000; oracle=pdp3-koblitz:m=2,engine=rootfind-negfold; solver=; solver_budget_seconds=0; targets=walk	–

Workloads

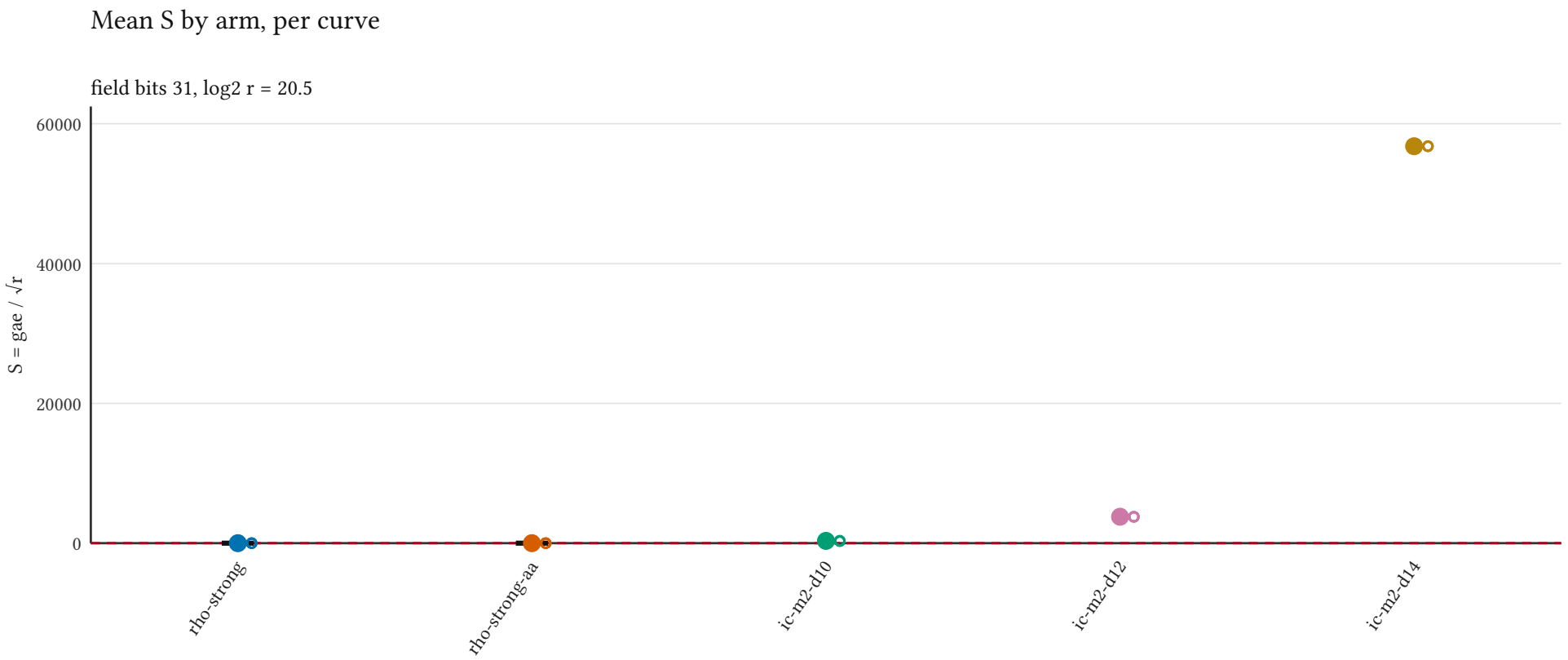
workload	curve	family	log2 r	A	target index	target seed
Wc179358520ce	icv1-f2m31-tm90707-c95f16f5	koblitz	20.457	62	0	20261002
Wec1f88aa247b	icv1-f2m31-tm90707-c95f16f5	koblitz	20.457	62	1	20261002
Wff09673378ab	icv1-f2m31-tm90707-c95f16f5	koblitz	20.457	62	2	20261002
Wa7cbaf1bca20	icv1-f2m31-tm90707-c95f16f5	koblitz	20.457	62	3	20261002

Runs

Planned 60 (20 warm-up, 40 measured); records 60 (40 measured); verified 40 (40 with matches_target); not verified 0; planned without a record 0.

No record has a status other than verified.

Graphs



— dot: mean S of the arm's verified runs on the curve; whisker: 95% bootstrap interval [d2]

- - dashed red: floor √(π/2A) [d3]

— black tick: expected S [d4]; hollow dot: per-workload mean

Session ECBS1h24a44d1bc573, n = 40 verified runs. Unit: S = gae / √r.

Figure 1: Mean S by arm per curve with 95% bootstrap whiskers, the floor and expected S; session ECBS1h24a44d1bc573, n = 40 verified runs, unit gae (S = gae / √r)

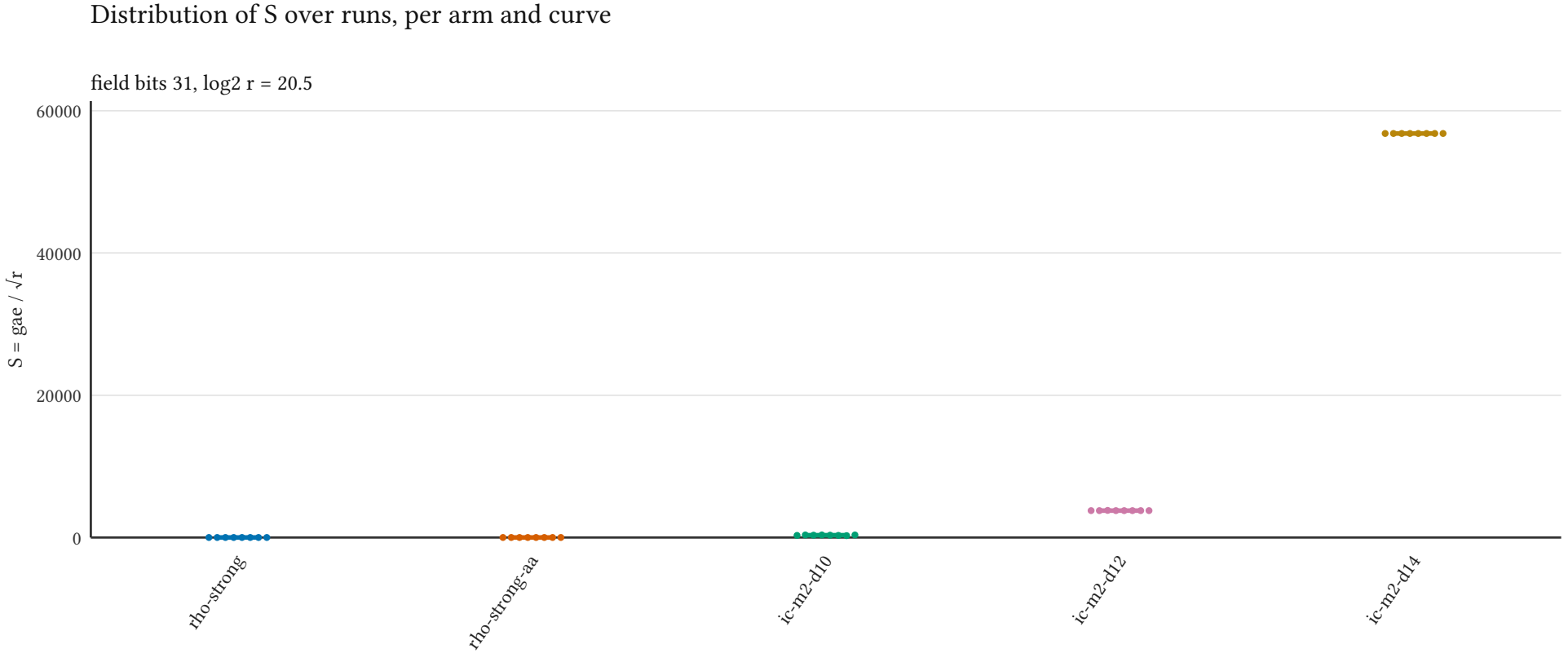


Figure 2: Distribution of S over the verified runs, per arm and curve; session ECBS1h24a44d1bc573, n = 40 verified runs, unit gae (S = gae / √r)

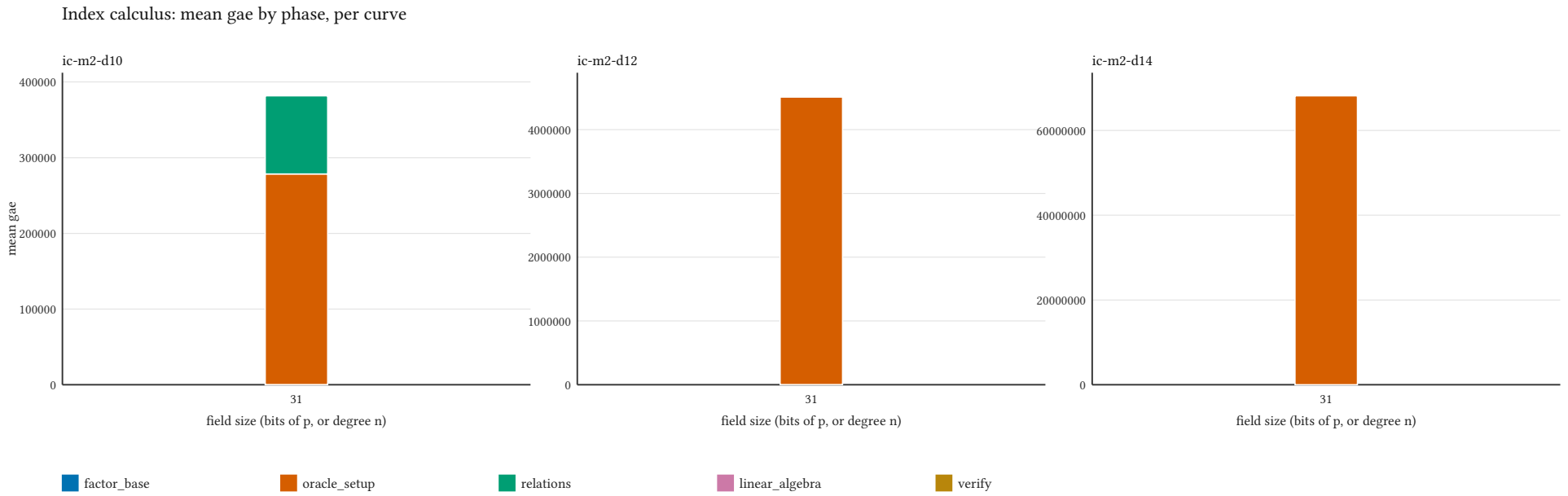


Figure 3: Mean gae by phase, index-calculus arms; session ECBS1h24a44d1bc573, n = 40 verified runs, unit gae (S = gae / √r)

Results per arm and curve

arm	curve	log2 r	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-strong	icv1-f2m31-tm90707-c95f16f5	20.457	8 / 8	1.2429	1.2294	1.0294	1.5637	0.1731	[1.1393, 1.3631]	0.1592	0.1592
rho-strong-aa	icv1-f2m31-tm90707-c95f16f5	20.457	8 / 8	1.2429	1.2294	1.0294	1.5637	0.1731	[1.1393, 1.3631]	0.1592	0.1592
ic-m2-d10	icv1-f2m31-tm90707-c95f16f5	20.457	8 / 8	318.1508	318.8477	269.6885	368.3744	38.0615	[293.9811, 343.1394]	0.1592	–
ic-m2-d12	icv1-f2m31-tm90707-c95f16f5	20.457	8 / 8	3775.7773	3773.1419	3763.3682	3798.8428	12.3065	[3768.3473, 3784.1511]	0.1592	–
ic-m2-d14	icv1-f2m31-tm90707-c95f16f5	20.457	8 / 8	56799.9846	56799.7501	56796.0884	56802.9644	2.1191	[56798.5610, 56801.3224]	0.1592	–

arm	curve	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-strong	icv1-f2m31-tm90707-c95f16f5	8 / 8	1491.1	7.8083	–	–	–	–	–	156.6037	189.6840	admitted (≥L0)
rho-strong-aa	icv1-f2m31-tm90707-c95f16f5	8 / 8	1491.1	7.8083	–	–	–	–	–	147.9350	175.7625	admitted (≥L0)
ic-m2-d10	icv1-f2m31-tm90707-c95f16f5	8 / 8	381700.5	1998.7981	–	–	–	–	–	115.7323	163.5370	admitted (≥L0)
ic-m2-d12	icv1-f2m31-tm90707-c95f16f5	8 / 8	4529977.7	23721.5069	–	–	–	–	–	1618.6592	1570.8855	admitted (≥L0)
ic-m2-d14	icv1-f2m31-tm90707-c95f16f5	8 / 8	68145614.4	356848.7044	–	–	–	–	–	47707.6351	47057.9180	admitted (≥L0)

Results per arm and workload

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-strong	Wc179358520ce	icv1-f2m31-tm90707-c95f16f5	2 / 2	1.3765	1.3765	1.1894	1.5637	0.2646	[1.1894, 1.5637]	0.1592	0.1592
rho-strong	Wec1f88aa247b	icv1-f2m31-tm90707-c95f16f5	2 / 2	1.1761	1.1761	1.0827	1.2694	0.1320	[1.0827, 1.2694]	0.1592	0.1592
rho-strong	Wff09673378ab	icv1-f2m31-tm90707-c95f16f5	2 / 2	1.0827	1.0827	1.0294	1.1361	0.0754	[1.0294, 1.1361]	0.1592	0.1592
rho-strong	Wa7cbaf1bca20	icv1-f2m31-tm90707-c95f16f5	2 / 2	1.3361	1.3361	1.2961	1.3761	0.0566	[1.2961, 1.3761]	0.1592	0.1592

arm	workload	curve	verified / planned	mean S [d1]	median S	min S	max S	stdev S	95% CI of mean [d2]	floor S [d3]	expected S [d4]
rho-strong-aa	Wc179358520ce	icv1-f2m31-tm90707-c95f16f5	2 / 2	1.3765	1.3765	1.1894	1.5637	0.2646	[1.1894, 1.5637]	0.1592	0.1592
rho-strong-aa	Wec1f88aa247b	icv1-f2m31-tm90707-c95f16f5	2 / 2	1.1761	1.1761	1.0827	1.2694	0.1320	[1.0827, 1.2694]	0.1592	0.1592
rho-strong-aa	Wff09673378ab	icv1-f2m31-tm90707-c95f16f5	2 / 2	1.0827	1.0827	1.0294	1.1361	0.0754	[1.0294, 1.1361]	0.1592	0.1592
rho-strong-aa	Wa7cbaf1bca20	icv1-f2m31-tm90707-c95f16f5	2 / 2	1.3361	1.3361	1.2961	1.3761	0.0566	[1.2961, 1.3761]	0.1592	0.1592
ic-m2-d10	Wc179358520ce	icv1-f2m31-tm90707-c95f16f5	2 / 2	299.9789	299.9789	281.9535	318.0043	25.4917	[281.9535, 318.0043]	0.1592	–
ic-m2-d10	Wec1f88aa247b	icv1-f2m31-tm90707-c95f16f5	2 / 2	324.3171	324.3171	280.2599	368.3744	62.3063	[280.2599, 368.3744]	0.1592	–
ic-m2-d10	Wff09673378ab	icv1-f2m31-tm90707-c95f16f5	2 / 2	294.6898	294.6898	269.6885	319.6912	35.3572	[269.6885, 319.6912]	0.1592	–
ic-m2-d10	Wa7cbaf1bca20	icv1-f2m31-tm90707-c95f16f5	2 / 2	353.6173	353.6173	351.9425	355.2921	2.3686	[351.9425, 355.2921]	0.1592	–
ic-m2-d12	Wc179358520ce	icv1-f2m31-tm90707-c95f16f5	2 / 2	3764.6776	3764.6776	3763.3682	3765.9869	1.8517	[3763.3682, 3765.9869]	0.1592	–
ic-m2-d12	Wec1f88aa247b	icv1-f2m31-tm90707-c95f16f5	2 / 2	3773.1189	3773.1189	3765.3729	3780.8649	10.9545	[3765.3729, 3780.8649]	0.1592	–
ic-m2-d12	Wff09673378ab	icv1-f2m31-tm90707-c95f16f5	2 / 2	3792.1707	3792.1707	3785.4985	3798.8428	9.4358	[3785.4985, 3798.8428]	0.1592	–
ic-m2-d12	Wa7cbaf1bca20	icv1-f2m31-tm90707-c95f16f5	2 / 2	3773.1419	3773.1419	3768.9132	3777.3707	5.9804	[3768.9132, 3777.3707]	0.1592	–
ic-m2-d14	Wc179358520ce	icv1-f2m31-tm90707-c95f16f5	2 / 2	56797.4447	56797.4447	56796.0884	56798.8009	1.9181	[56796.0884, 56798.8009]	0.1592	–
ic-m2-d14	Wec1f88aa247b	icv1-f2m31-tm90707-c95f16f5	2 / 2	56800.5538	56800.5538	56799.3464	56801.7611	1.7075	[56799.3464, 56801.7611]	0.1592	–
ic-m2-d14	Wff09673378ab	icv1-f2m31-tm90707-c95f16f5	2 / 2	56800.7603	56800.7603	56800.1049	56801.4156	0.9268	[56800.1049, 56801.4156]	0.1592	–
ic-m2-d14	Wa7cbaf1bca20	icv1-f2m31-tm90707-c95f16f5	2 / 2	56801.1798	56801.1798	56799.3952	56802.9644	2.5237	[56799.3952, 56802.9644]	0.1592	–

arm	workload	matches target	mean gae [d3]	S / floor [d3]	dp [d6]	steps [d6]	muls [d6]	sqrs [d6]	invs [d6]	wall ms [d7]	cpu ms [d7]	time status [d7]
rho-strong	Wc179358520ce	2 / 2	1651.5	8.6482	–	–	–	–	–	137.5330	168.4170	admitted (≥L0)
rho-strong	Wec1f88aa247b	2 / 2	1411.0	7.3888	–	–	–	–	–	193.6931	226.1710	admitted (≥L0)
rho-strong	Wff09673378ab	2 / 2	1299.0	6.8023	–	–	–	–	–	184.5721	211.5960	admitted (≥L0)
rho-strong	Wa7cbaf1bca20	2 / 2	1603.0	8.3942	–	–	–	–	–	140.6978	164.6180	admitted (≥L0)
rho-strong-aa	Wc179358520ce	2 / 2	1651.5	8.6482	–	–	–	–	–	160.1174	175.3915	admitted (≥L0)
rho-strong-aa	Wec1f88aa247b	2 / 2	1411.0	7.3888	–	–	–	–	–	147.2187	175.7625	admitted (≥L0)
rho-strong-aa	Wff09673378ab	2 / 2	1299.0	6.8023	–	–	–	–	–	148.5387	184.1610	admitted (≥L0)
rho-strong-aa	Wa7cbaf1bca20	2 / 2	1603.0	8.3942	–	–	–	–	–	142.6213	171.7380	admitted (≥L0)
ic-m2-d10	Wc179358520ce	2 / 2	359898.8	1884.6322	–	–	–	–	–	111.8775	153.7125	admitted (≥L0)
ic-m2-d10	Wec1f88aa247b	2 / 2	389098.5	2037.5385	–	–	–	–	–	91.7563	135.1795	admitted (≥L0)
ic-m2-d10	Wff09673378ab	2 / 2	353553.2	1851.4033	–	–	–	–	–	105.8524	152.1320	admitted (≥L0)
ic-m2-d10	Wa7cbaf1bca20	2 / 2	424251.3	2221.6182	–	–	–	–	–	141.8315	190.6635	admitted (≥L0)
ic-m2-d12	Wc179358520ce	2 / 2	4516660.8	23651.7725	–	–	–	–	–	1561.8974	1574.6150	admitted (≥L0)
ic-m2-d12	Wec1f88aa247b	2 / 2	4526788.3	23704.8056	–	–	–	–	–	1493.9064	1528.3495	admitted (≥L0)
ic-m2-d12	Wff09673378ab	2 / 2	4549645.6	23824.4991	–	–	–	–	–	1513.1554	1527.5365	admitted (≥L0)
ic-m2-d12	Wa7cbaf1bca20	2 / 2	4526815.9	23704.9503	–	–	–	–	–	1692.7430	1647.9305	admitted (≥L0)
ic-m2-d14	Wc179358520ce	2 / 2	68142567.1	356832.7470	–	–	–	–	–	47643.5714	46162.6890	admitted (≥L0)
ic-m2-d14	Wec1f88aa247b	2 / 2	68146297.2	356852.2800	–	–	–	–	–	47360.2649	46768.9795	admitted (≥L0)
ic-m2-d14	Wff09673378ab	2 / 2	68146544.9	356853.5774	–	–	–	–	–	46710.7552	45826.5190	admitted (≥L0)
ic-m2-d14	Wa7cbaf1bca20	2 / 2	68147048.3	356856.2132	–	–	–	–	–	48341.3302	47624.9635	admitted (≥L0)

Phases of the index-calculus arms

arm	curve	phase	runs	mean gae	share [d8]
ic-m2-d10	icv1-f2m31-tm90707-c95f16f5	factor_base	8	112.5	0.0%
ic-m2-d10	icv1-f2m31-tm90707-c95f16f5	oracle_setup	8	278255.0	72.9%
ic-m2-d10	icv1-f2m31-tm90707-c95f16f5	relations	8	103304.9	27.1%
ic-m2-d10	icv1-f2m31-tm90707-c95f16f5	linear_algebra	8	1.1	0.0%
ic-m2-d10	icv1-f2m31-tm90707-c95f16f5	verify	8	27.0	0.0%

arm	curve	phase	runs	mean gae	share [d8]
ic-m2-d12	icv1-f2m31-tm90707-c95f16f5	factor_base	8	450.9	0.0%
ic-m2-d12	icv1-f2m31-tm90707-c95f16f5	oracle_setup	8	4509251.0	99.5%
ic-m2-d12	icv1-f2m31-tm90707-c95f16f5	relations	8	20245.4	0.4%
ic-m2-d12	icv1-f2m31-tm90707-c95f16f5	linear_algebra	8	3.3	0.0%
ic-m2-d12	icv1-f2m31-tm90707-c95f16f5	verify	8	27.0	0.0%
ic-m2-d14	icv1-f2m31-tm90707-c95f16f5	factor_base	8	1789.0	0.0%
ic-m2-d14	icv1-f2m31-tm90707-c95f16f5	oracle_setup	8	68136769.0	100.0%
ic-m2-d14	icv1-f2m31-tm90707-c95f16f5	relations	8	7016.0	0.0%
ic-m2-d14	icv1-f2m31-tm90707-c95f16f5	linear_algebra	8	13.4	0.0%
ic-m2-d14	icv1-f2m31-tm90707-c95f16f5	verify	8	27.0	0.0%

Raw records

`research\/koblitz\_halfdim\_base\_20261018\/sessions\/n31/records.jsonl` (60 records; ids in `raw_records.csv` and `report.json`); child stderr in `research\/koblitz\_halfdim\_base\_20261018\/sessions\/n31/exec/` (0 file(s)).

Derivations

- **[d1]** S = cost.s of each verified, non-warm-up record (`total_gae / √r`, stored by the runner); mean, median, min, max and sample stdev (stats.rs) are over those records.
- **[d2]** 95% percentile bootstrap interval of the mean S: 4000 resamples of the verified runs of the cell with replacement (stats::bootstrap_ci, seed 20261002). A cell with one run has no interval. Not a significance test.
- **[d3]** total gae and S / floor are means over the verified runs of the cell of cost.total_gae and boundaries.ratio_to_floor; floor S is boundaries.floor_s = √(π / 2A), A = automorphisms the curve offers a generic method.
- **[d4]** expected S is methods::expected_s(method, A): the closed form the method's mean S approaches as r grows (set-up omitted). A reference mark, not a measurement.
- **[d5]** scaling: per arm, per curve, the mean gae over the arm's verified runs on that curve; the line is ordinary least squares of log2(mean gae) on log2(r) over the curves (bounds::ols), slope α, intercept log2 C so gae = C·r^α. The standard error of α is √(SS_res/(sizes−2)/S_xx) from those residuals (none below 3 sizes). Counted operations, one point per size: it is not the cluster-bootstrap interval of `ecbench bound fit`, which resamples runs and sizes.
- **[d6]** counters (distinguished points, steps) and field operations (muls, sqrs, invs) are means over the cell's verified runs of the counters the record carries (steps = `steps`, or baby + giant steps for BSGS); a dash means the method did not count it (unknown, not zero).
- **[d7]** wall and CPU are medians over the cell's verified runs of time.solve_wall_ns and time.user_ns + sys_ns. The figure is admitted only when every run behind it earned the level the spec requires (compare.rs's gate); otherwise it is informative at the lowest level earned, and never a result.
- **[d8]** phase share = the phase's mean gae over the sum of the arm's mean gae across phases on that curve, over verified runs.